

Appendix O

Graduate-Level AI in Education Course Modules

Natural Language Processing in Education and Student Use of AI in Learning

Graduate-Level AI in Education Course Modules

Developed for graduate course design, August 2024

Module 1: Natural Language Processing (NLP) in Education

Title: Leveraging Natural Language Processing to Enhance Educational Outcomes

Overview

This module introduces the fundamental concepts of Natural Language Processing (NLP) and its application in educational contexts. Participants will explore various NLP tools, such as chatbots, automated essay scoring systems, and translation services, and learn how to integrate these technologies into teaching practices. The module will also cover the ethical considerations and challenges associated with the use of NLP in education.

Objectives

1. Define key terms and concepts related to NLP, including tokenization, sentiment analysis, and machine learning.
2. Explore the functionality and potential applications of NLP tools in educational settings.
3. Evaluate the impact of NLP on student learning, particularly in areas like writing and language development.
4. Discuss ethical considerations related to the use of NLP technologies, including privacy concerns and the potential for bias.
5. Develop a plan to integrate NLP tools into curriculum design and student assessment.

Required Learning Materials

- Article: "ChatGPT as an Automated Essay Scoring Tool in the Writing Classroom: How It Compares with Human Scoring." Compares AI-based essay scoring and human assessment, highlighting the potential and limitations of NLP tools in education.
- Article: "Interacting with Educational Chatbots: A Systematic Review." Explores the roles chatbots can play in educational settings, such as teaching agents and motivational agents, and the implications for student engagement and learning.
- Video: "How Chatbots are Changing Education." Demonstrates practical applications of chatbots in education with real-world examples.
- Case study: "Towards Automated Writing Evaluation: A Comprehensive Review." Examines the development and use of automated writing evaluation systems such as Grammarly and Criterion.

Optional Materials

- Journal article: Ethical Considerations in the Development of NLP Technologies, focusing on bias and data privacy.

Practice Activities

- Tool exploration: participants use NLP tools, including automated essay scorers and educational chatbots, to assess sample student essays and engage in conversations with chatbots to understand their educational applications.
- Discussion forum: participants discuss the implications of using NLP in their specific teaching contexts, addressing both benefits and drawbacks. Prompt: How do the tools discussed impact your current or future teaching practices?

Graded Activities

- Critical evaluation essay (3 to 5 pages): evaluate a specific NLP tool, discussing its functionality, benefits, challenges, and classroom integration, including a section on ethical considerations.
- Project: develop a lesson plan or unit incorporating NLP tools within the content area the participant currently teaches, demonstrating thoughtful integration of the technology for learning and assessment.

Project Evaluation Checklist

- Objectives (20 points): clearly defined learning objectives that align with the use of NLP tools.
- Instructional strategies (30 points): effective and innovative use of NLP tools in the lesson plan.
- Integration of NLP tools (30 points): tools well integrated into the curriculum, with specific examples of application.
- Assessment plan (20 points): a clear and effective plan for assessing student learning and the impact of NLP tools.

Rubrics: separate rubrics were developed for the evaluation essay and the lesson plan project.

Module 2: Student Use of AI in Education

Title: Empowering Students through the Use of AI Tools in Learning

Overview

This module explores how AI tools can be effectively used by students to enhance their learning and engagement. The focus is on practical applications of AI in areas such as writing, brainstorming, idea development, and simulation-based learning. Participants also examine how AI can foster metacognition and deeper learning, and how it affects student motivation.

Objectives

1. Identify and describe AI tools that can support student learning, such as writing assistants, brainstorming aids, and AI-driven simulations.
2. Analyze the impact of AI tools on student motivation, engagement, and learning outcomes.
3. Design activities that effectively incorporate AI tools to enhance student learning and promote metacognition.
4. Reflect on the ethical and pedagogical implications of AI use in student learning.

Required Learning Materials

- Article: "AI-Generated Feedback on Writing: Insights into Efficacy." Examines the impact of AI-generated feedback on student writing, focusing on engagement and learning outcomes.
- Video tutorial: "Using AI Tools to Support Student Writing." Shows how tools such as Grammarly can be used to improve student writing skills and engagement.

Optional Materials

- Podcast: "AI in Education: Student Perspectives," featuring insights from students who have used AI tools in their learning.
- Webinar: "Harnessing AI for Student Success," on strategies for using AI to enhance student learning and success.

Practice Activities

- Tool usage practice: participants use AI tools such as Grammarly and Jasper for writing support and brainstorming, screen record the experience, and submit a clip to discuss in their reflection.
- Collaborative discussion on how AI can foster deeper learning and metacognition, including challenges and best practices. Prompt: How does the evidence presented in "AI-Generated Feedback on Writing" influence your perspective on using AI tools in your teaching?

Graded Activities

- Concept mapping: use AI tools to create a detailed concept map that organizes and connects ideas related to a specific course topic, building understanding of complex material.

A rubric was developed for the concept mapping assignment.